

2.4 Diffusion on the whole line

Consider the following diffusion equation

$$\begin{aligned} u_t &= ku_{xx}, \quad (x \in \mathbb{R}, t > 0) \\ u(x, 0) &= \phi(x). \end{aligned}$$

To construct a solution, one needs to investigate its *invariance properties*.

- (a) A translation $u_y(x, t) := u(x - y, t)$ of any solution $u(x, t)$ is another solution.¹ for any fixed $y \in \mathbb{R}$.
- (b) Any derivative of a solution is again a solution.
- (c) A linear combination of solutions is again a solution.
- (d) An integral of a solution², is again a solution.
- (e) If $u(x, t)$ is a solution, so is the dilated function $u(\sqrt{a}x, at)$ for any $a > 0$.

Remark. 1. (a) + (d) : If $S(x, t)$ solves the equation, so does $S(x - y, t)$ and

$$v(x, t) := \int_{-\infty}^{\infty} S(x - y, t)g(y) dy$$

as long as the integral is justified.

- 2. Note that for (e), we seek a special solution $Q(x, t)$ which is dilation-invariant. Choose an initial data for Q as

$$Q = \begin{cases} Q(x, 0) = 0, & x < 0 \\ Q(x, 0) = 1, & x > 0 \end{cases}$$

Step 1. We even look for $Q(x, t)$ if the special form

$$Q(x, t) = g(p), \quad p = \frac{x}{\sqrt{4kt}}. \quad (*)$$

so that we can obtain a simpler formula for Q .

Step 2. Put $(*)$ into the diffusion equation. Then we have

$$\begin{aligned} Q_t &= \frac{dg}{dp} \cdot \frac{dp}{dt} = g'(p) \cdot p_t = g'(p) \cdot \left(-\frac{x}{2\sqrt{4kt^3}} \right) = -\frac{p}{2t} g'(p), \\ Q_{xx} &= \frac{dg}{dp} \cdot \frac{dp}{dx} = g'(p) \cdot p_x = g'(p) \cdot \frac{1}{\sqrt{4kt}} = g''(p) \cdot p_x^2 = g''(p) \cdot \frac{1}{4kt}. \end{aligned}$$

Thus, the diffusion equation becomes

$$0 = Q_t - kQ_{xx} = \frac{1}{t} \left(-\frac{p}{2} g'(p) - \frac{1}{4} g''(p) \right)$$

which is equivalent to

$$\frac{1}{4} g''(p) + \frac{1}{2} p g'(p) = 0 \iff g''(p) + 2p g'(p) = 0 \iff (e^{p^2} g'(p))' = 0.$$

¹We disregard the initial condition for now.

²For example, $\int_0^x u(y, t) dy$

Hence, we have

$$g(p) = c_1 \int_0^p e^{-s^2} ds + c_2.$$

So we set

$$Q(x, t) = c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2.$$

Step 3. Due to the initial condition, if $x > 0$,

$$1 = \lim_{t \searrow 0} Q(x, t) = \lim_{t \searrow 0} \left(c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2 \right) = c_1 \int_0^\infty e^{-p^2} dp + c_2 = c_1 \frac{\sqrt{\pi}}{2} + c_2$$

and if $x < 0$,

$$0 = \lim_{t \searrow 0} Q(x, t) = \lim_{t \searrow 0} \left(c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2 \right) = c_1 \int_0^{-\infty} e^{-p^2} dp + c_2 = -c_1 \frac{\sqrt{\pi}}{2} + c_2.$$

Thus, we can find that

$$c_1 = \frac{1}{\sqrt{\pi}}, \quad c_2 = \frac{1}{2}.$$

Hence, if we set

$$Q(x, t) := \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp$$

then, we set

$$u(x, t) := \int_{-\infty}^\infty S(x - y, t) \phi(y) dy$$

where

$$S(x, t) := Q_x(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}$$

and $\phi(x)$ is the initial data. Since $\frac{\partial Q}{\partial x}$ solves the equation, u satisfies

$$u_t = k u_{xx}.$$

To check $\lim_{t \searrow 0} u(x, t) = \phi(x)$, we assume that

$$\lim_{|y| \rightarrow \infty} \phi(y) = 0, \quad \phi : \text{ is differentiable } \quad \text{and} \quad \int_{-\infty}^\infty |\phi'(y)| dy < \infty.$$

Then one can check that

$$\begin{aligned} u(x, t) &= - \int_{-\infty}^\infty S(x - y, t) \phi(y) dy \\ &= - \int_{-\infty}^\infty \frac{\partial}{\partial t} y \{ Q(x - y, t) \} \phi(y) dy \\ &= - [Q(x - y, t) \phi(y)]_{y=-\infty}^{y=\infty} + \int_{-\infty}^\infty Q(x - y, t) \phi'(y) dy \\ &= \int_{-\infty}^\infty Q(x - y, t) \phi'(y) dy. \end{aligned}$$

Therefore, we have

$$\begin{aligned}
\lim_{t \searrow 0} u(x, t) &= \lim_{t \searrow 0} \int_{-\infty}^{\infty} Q(x - y, t) \phi'(y) dy \\
&= \int_{-\infty}^{\infty} \lim_{t \searrow 0} Q(x - y, t) \phi'(y) dy \\
&= \int_{-\infty}^x \phi'(y) dy \\
&= \phi(x).
\end{aligned} \tag{**}$$

Note that we used the dominated convergence theorem in (**), and we get the pointwise convergence of the solution to the initial data.

Now we have the following formula:

$$u(x, t) = \int_{-\infty}^{\infty} S(x - y, t) \phi(y) dy = S(\cdot, t) * \phi(x)$$

Such function S is called the source function, Green's function, fundamental solution, or the heat kernel.

Remark.

1. The source function, if regarded as a family of functions parametrized by t ,³ forms a family of good kernels as $t \searrow 0$, that is, it satisfies the following properties:

- (i) $\int_{-\infty}^{\infty} S(x, t) dx = 1$ for all $t > 0$.
- (ii) $\sup_{t > 0} \int_{-\infty}^{\infty} |S(x, t)| dx < \infty$.
- (iii) $\int_{|x| > \delta} S(x, t) dx \rightarrow 0$ as $t \searrow 0$ for each $\delta > 0$.

Moreover,

$$\int_{-\infty}^{\infty} |\phi(x)| dx < \infty \implies \int_{-\infty}^{\infty} |S(\cdot, t) * \phi - \phi| dx \rightarrow 0 \text{ as } t \searrow 0$$

which means the L^1 convergence of the solution formula to the initial data.

2. As $t \rightarrow \infty$,

$$S(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}} \rightarrow 0 \quad \text{uniformly in } x \in \mathbb{R}.$$

since $|S(x, t)| \leq \frac{1}{\sqrt{4\pi kt}}$. Thus, once ϕ is merely integrable, then

$$u := S(\cdot, t) * \phi$$

satisfies the heat equation on $(-\infty, \infty) \times (0, \infty)$ and becomes a smooth function.

3. Reflections and Sources

Diffusion on the half line (Sec. 3.1 and 3.3)

Now we consider the homogeneous diffusion equation with homogeneous Dirichlet boundary condition on the half line:

$$\begin{aligned}
v_t &= kv_{xx}, & (x > 0, t > 0) \\
v(x, 0) &= \phi(x), & (x > 0) \\
v(0, t) &= 0, & (t > 0)
\end{aligned}$$

³Not as $S(x, t)$, but as $\{S(\cdot, t)\}_{t > 0}$.

Our strategy is to extend our problem to the whole real line and restrict the solution on $\mathbb{R}$ to $[0, \infty)$. For this, we consider the odd extension ϕ_{odd} of the initial data ϕ ;

$$\phi_{\text{odd}}(x) := \begin{cases} \phi(x), & x > 0 \\ 0, & x = 0 \\ -\phi(-x), & x < 0 \end{cases}$$

Let $u(x, t)$ be the solution of

$$\begin{aligned} u_t &= ku_{xx}, & (x \in \mathbb{R}, t > 0) \\ u(x, 0) &= \phi_{\text{odd}}(x), & (x \in \mathbb{R}) \end{aligned}$$

then we have

$$u(x, t) = S(\cdot, t) * \phi_{\text{odd}}(x) = \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(y) dy$$

solves the equation.

Our choice is to take

$$v(x, t) := u(x, t)|_{x \geq 0}.$$

One needs to check $u(0, t) = 0$; First, u is smooth once ϕ is integrable.⁴ Furthermore,

$$\begin{aligned} u(-x, t) &= \int_{-\infty}^{\infty} S(-x - y, t) \phi_{\text{odd}}(y) dy \\ &= \int_{-\infty}^{\infty} S(x + y, t) \phi_{\text{odd}}(y) dy \\ &= \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(-y) dy \\ &= - \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(y) dy \\ &= -u(x, t) \end{aligned}$$

recall that S is an even function. Thus, $v(0, t) = u(0, t) = 0$ for all $t > 0$.

If ϕ satisfies good properties, then

$$\lim_{t \searrow 0} v(x, t) = \phi(x)$$

and we can see

$$\begin{aligned} v(x, t) &= \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(y) dy \\ &= \int_0^{\infty} S(x - y, t) \phi(y) dy - \int_{-\infty}^0 S(x - y, t) \phi(y) dy \\ &= \int_0^{\infty} (S(x - y, t) - S(x + y, t)) \phi(y) dy. \end{aligned}$$

This kind of method is called the **method of odd extensions** or the **reflection method**.

⁴This is basic assumption for initial data of the diffusion equation.